WORKED SOLUTION

STEP II 1995, Question 1

Pure • Differentiating Geometric Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q1

The key idea

A finite geometric series has a closed form, and an identity between two functions of x can be differentiated to give a new identity. Differentiating $\sum x^k$ brings down a factor k, which is exactly what turns a geometric series into $\sum kx^{k-1}$; putting $x = -1$ then supplies the alternating signs. For the squares, the cleanest route is to let the answer's shape tell us what A and B must be, then prove it by induction.

Solution

Part (i)

Write $S = 1 + x + x^2 + \cdots + x^n$. Then

$$S(1-x) = (1 + x + \cdots + x^n) - (x + x^2 + \cdots + x^{n+1}) = 1 - x^{n+1},$$

since every power $x, x^2, \ldots, x^n$ appears once with each sign and cancels. If $x \neq 1$ then $1 - x \neq 0$, so we may divide by it:

$$1 + x + x^2 + \cdots + x^n = \frac{1 - x^{n+1}}{1 - x},$$

as required.

Part (ii)

Both sides of (i) are differentiable functions of x on $x \neq 1$ and agree there, so their derivatives agree. On the left,

$$\frac{\mathrm{d}}{\mathrm{d}x}(1 + x + \cdots + x^n) = 1 + 2x + 3x^2 + \cdots + nx^{n-1}.$$

On the right, by the quotient rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1 - x^{n+1}}{1 - x}\right) = \frac{-(n+1)x^n(1-x) + (1 - x^{n+1})}{(1-x)^2} = \frac{1 - (n+1)x^n + nx^{n+1}}{(1-x)^2}.$$

Now set $x = -1$, which is allowed because $-1 \neq 1$. The left-hand side becomes $1 - 2 + 3 - 4 + \cdots + (-1)^{n-1}n$, because the term kx^{k-1} becomes $(-1)^{k-1}k$. The right-hand side becomes

$$\frac{1 - (n+1)(-1)^n + n(-1)^{n+1}}{4} = \frac{1 - (-1)^n(2n+1)}{4},$$

using $(-1)^{n+1} = -(-1)^n$.

- If n is even, $(-1)^n = 1$ and the value is $\dfrac{1 - 2n - 1}{4} = -\dfrac{n}{2}$.
- If n is odd, $(-1)^n = -1$ and the value is $\dfrac{1 + 2n + 1}{4} = \dfrac{n+1}{2}$.

This is what we had to show.

Part (iii)

Let $T_n = 1^2 - 2^2 + 3^2 - \cdots + (-1)^{n-1}n^2$, and suppose $T_n = (-1)^{n-1}f(n)$ with $f(n) = An^2 + Bn$.

Ask yourself How can we find A and B before we know the formula is true?

Use the one fact we know for certain about T_n: consecutive sums differ by the last term, $T_n - T_{n-1} = (-1)^{n-1}n^2$. Since $T_{n-1} = (-1)^{n-2}f(n-1) = -(-1)^{n-1}f(n-1)$, this reads

$$(-1)^{n-1}\big(f(n) + f(n-1)\big) = (-1)^{n-1}n^2, \qquad \text{so} \qquad f(n) + f(n-1) = n^2.$$

With $f(n) = An^2 + Bn$,

$$\begin{aligned} f(n) + f(n-1) &= An^2 + Bn + A(n-1)^2 + B(n-1) \\ &= 2An^2 + (2B - 2A)n + (A - B). \end{aligned}$$

For this to equal n^2 for every n we need $2A = 1$ and $B = A$, that is

$$A = B = \tfrac{1}{2}, \qquad T_n = (-1)^{n-1}\,\frac{n(n+1)}{2}.$$

(These values are forced: $n = 1$ gives $A + B = 1$ and $n = 2$ gives $-(4A + 2B) = -3$, whose only solution is $A = B = \frac{1}{2}$.)

Proof by induction. For $n = 1$, $T_1 = 1 = (-1)^0 \cdot \frac{1\cdot 2}{2}$. Suppose $T_{n-1} = (-1)^{n-2}\frac{(n-1)n}{2}$ for some $n \geq 2$. Then

$$\begin{aligned} T_n = T_{n-1} + (-1)^{n-1}n^2 &= (-1)^{n-1}\left(n^2 - \frac{n(n-1)}{2}\right) \\ &= (-1)^{n-1}\,\frac{n(n+1)}{2}, \end{aligned}$$

which is the formula for n. So it holds for all $n \geq 1$, with $A = B = \frac{1}{2}$.

Alternatively, pair the terms: $(2m-1)^2-(2m)^2=-(4m-1)$. For $n=2m$ this gives $T_n=-\sum_{j=1}^{m}(4j-1)=-m(2m+1)=-\frac{n(n+1)}{2}$; for $n=2m+1$ add $(2m+1)^2$ to get $(2m+1)(m+1)=\frac{n(n+1)}{2}$. (One could also multiply the identity of (ii) by x and differentiate again, but the algebra is much heavier.)

Checks

For $n=4$: $1-2+3-4=-2=-\frac{4}{2}$, and $1-4+9-16=-10=-\frac{4\cdot5}{2}$. For $n=5$: $1-2+3-4+5=3=\frac{6}{2}$, and $-10+25=15=\frac{5\cdot6}{2}$. A computer check of all three identities for n up to 500 (for example $T_{10}=-55$ and $T_{11}=66$) agrees exactly.

Notice also that $|T_n|=1+2+\cdots+n$: the alternating sum of squares has the same size as the plain sum of the first n integers, a surprising coincidence that the pairing argument explains.

Common slips

- Forgetting that $x=-1$ is a legitimate value only because the identity in (i) holds for all $x\neq1$; at $x=1$ the right-hand side is undefined.
- Sign errors with $(-1)^{n+1}$: it equals $-(-1)^n$, not $(-1)^n$.
- Finding A and B from $n=1$ and $n=2$ and stopping there. That only shows these are the *only possible* values; the induction (or pairing) shows the formula actually holds for every n.

Beyond STEP

Differentiating $\sum x^k$ term by term is the finite version of a powerful idea: power series can be differentiated term by term inside their radius of convergence, giving $\sum kx^{k-1}=(1-x)^{-2}$ for $|x|<1$. At $x=-1$ the infinite series $1-2+3-4+\cdots$ diverges, yet the formula suggests the value $\frac{1}{4}$, and indeed the formula in (ii) can be written $\frac{1}{4}-(-1)^n\frac{2n+1}{4}$, so the partial sums swing symmetrically about $\frac{1}{4}$. Giving such divergent series a value is the idea behind Abel summation. Abbott's *Understanding Analysis* treats term-by-term differentiation of power series carefully.

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